

Extraction of Cardiac Time Intervals From a Radar Kinetocardiogram for Continuous Blood Pressure Monitoring

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Abstract—Noncontact blood pressure (BP) monitoring remains challenging. Conventional pulse-wave-based approaches rely on fine-grained waveform morphology and are vulnerable to respiratory interference. Deep learning can improve accuracy, yet often lacks physiological interpretability and requires large training datasets. This work proposes a radar-kinetocardiogram (RKG)-based BP monitoring method that does not rely on deep learning. To suppress respiratory interference, we introduce Adaptive Respiratory-Harmonic Variational Mode Extraction (ARH-VME) effectively removes respiratory fundamentals and harmonics while preserving key cardiac components. By extracting cardiac time intervals (CTIs) from the RKG and estimating BP primarily from these temporal markers, the method markedly reduces dependence on detailed waveform morphology and improves robustness. In a clinical study with 25 subjects, the approach achieved a mean error (ME), mean absolute error (MAE), and standard deviation (SD) of $0.3 (5.72) \pm 6.44$ mmHg for systolic BP (SBP) and $-0.4 (3.02) \pm 3.6$ mmHg for diastolic BP (DBP) under static conditions, falling within the numerical limits commonly referenced from the AAMI standard and demonstrated stable performance over 40–100 cm sensing distances as well as incidence angles up to 45° . Cross-platform generalizability was further validated on an independent public dataset. These results indicate a physiologically interpretable and robust pathway for practical, noncontact BP monitoring.

Index Terms—Adaptive respiratory-harmonic variational mode extraction (ARH-VME), cardiac time intervals (CTIs), frequency-modulated continuous wave, noncontact blood pressure (BP) monitoring, radar kinetocardiogram (RKG).

I. INTRODUCTION

CARDIOVASCULAR disease remains the leading cause of mortality, and hypertension is increasingly affecting younger populations. These trends highlight the need for continuous blood pressure (BP) monitoring to maintain cardiovascular health and prevent disease progression [1].

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Conventionally, BP is monitored using cuff-based sphygmomanometers, but sustained inflation causes discomfort and limits long-duration monitoring [2]. Invasive arterial catheterization provides high accuracy yet carries infection risk and is unsuitable for routine use [3]. Noninvasive finger-cuff devices offer continuous readings but remain contact-based, require calibration, and are relatively costly [4]. Most other techniques provide only intermittent measurements. There is therefore an urgent need for a noncontact modality that enables continuous BP monitoring with adequate accuracy and privacy.

In recent years, FMCW radar has seen rapid progress in noncontact sensing of human physiological signals. By analyzing body-reflected echoes, such radar can extract respiration, heartbeat, and BP without skin contact, which suits burn patients, intensive care units (ICUs) settings, and infant monitoring while posing minimal privacy concerns. Current radar-based BP estimation methods can be broadly divided into pulse transit time (PTT)-based approaches and deep learning regression approaches. PTT is the arterial pulse propagation delay between two sites; as BP rises, pulse wave velocity (PWV) increases and PTT decreases, enabling BP estimation from either quantity [5].

In radar sensing, researchers aim to remove the dependence on ECG by measuring arterial pulse waves at one or more body sites, thereby deriving PTT, PWV, and related indices. Wen et al. [6] integrated a dual-frequency radar system for simultaneous chest-motion and pulse detection, enabling monitoring of heart-rate variability (HRV), PTT, PWV, and BP. Tseng and Tseng [7] used a 24 GHz continuous-wave radar and extracted the reflective PTT (rPTT). Unlike conventional PTT, rPTT was redefined as the propagation time interval between the forward and reflected pressure waves in the radial artery region. Zhao et al. [8] captured the tiny surface displacements of the central aorta using radar and extracted carotid-femoral PTT (cf-PTT) for BP calculation. Geng et al. [9] used a single millimeter-wave radar to obtain central-artery PTT (caPTT) from both the neck and chest. They defined the time difference between the chest and neck main pulse peaks as the radar-detected transit (RDT) time, which showed over 90 % correlation with PTT. These studies have demonstrated the feasibility of radar-based PTT estimation, but several limitations remain. During experiments, participants are typically required to hold their breath, suggesting that

the measured pulse waveform is vulnerable to respiratory interference, which may cause PTT extraction to fail and degrade BP accuracy. To address this, Zheng et al. [10] proposed adaptive respiratory harmonic filtering (ARHF) to suppress respiratory interference on the Doppler wrist pulse (DWP), and extracted features such as rPTT and diameter variation ratio (DVR) for multidimensional morphological BP regression. However, when the measurement distance exceeds a certain range, the DWP waveform becomes difficult to detect, limiting BP accuracy. Overall, these PTT-based methods require subject-specific calibration and remain sensitive to respiratory interference and measurement distance.

With the continuous development of deep learning and machine learning, many BP estimation models that do not require calibration have emerged. Qiu et al. [11] measured the chest pulse microvibrations (CPMVs) using radar and demonstrated the high correlation between CPMV and PPG signals. They replaced the PPG input with CPMV in a stacked deformable convolution network (RSD-Net) that integrated channel and spatial self-attention mechanisms for continuous BP estimation. Wang et al. [12] captured tiny chest wall vibrations and their second derivatives using millimeter-wave radar and input the two channels into a hierarchical CNN-Transformer network, enabling BP estimation every 2 s. Those authors recognized the need to find a signal easier to obtain than pulse waves, though the chest microvibration signals they extracted still required high waveform granularity. Wang et al. [13] collected phase signals from the neck using FMCW radar and applied time-varying-filter-based EMD to decompose the phase signal. The reconstructed pulse wave was input into a dual-stream fusion neural network (DSFNN-BP) for BP estimation. Liang et al. [14] captured wrist arterial pulse signals using millimeter-wave radar and built AirBP, which established a BP perception pretraining model with a multiattention architecture to extract effective features from sparse and noisy radar signals. Shi et al. [15] automatically extracted arterial pulse wave features from radar echoes and proposed a random-forest-based BP measurement model. They further introduced a Delay-Doppler domain feature transformation method to improve pulse wave signal quality and compared three machine learning methods for BP regression [16]. Zheng et al. [17] discovered that individuals with drastically different BP values may exhibit very similar pulse waveforms, leading to confusion in model learning. They integrated physiological factors into a data-driven deep learning framework, and experimental results showed that the proposed BP3 model significantly improved performance for subjects with abnormal BP.

Although the aforementioned methods have addressed some challenges in noncontact radar-based BP monitoring, fundamental issues remain. On one hand, PTT extraction requires high-fidelity pulse waveforms, but pulse wave measurement is easily interfered with by respiration, body movement, and background noise. In clinical experiments, it is nearly impossible to consistently obtain clean pulse waveforms at all times. Some researchers have discarded partially distorted waveforms to improve accuracy, but this sacrifices the BP sampling rate [18]. On the other hand, deep learning models lack

physiological interpretability due to their complex structure, which affects clinical trust and poses obstacles to regulatory acceptance. Moreover, training with noise-contaminated pulse waves wastes computational resources and may degrade model accuracy.

The theory of estimating BP based on key time points of the cardiac time intervals (CTIs) provides a promising new perspective for noncontact continuous BP monitoring [19]. This theory does not directly rely on the complete and detailed shape of the pulse wave but focuses on time markers directly determined by cardiac mechanical activity. However, the use of heart rate phase signals is susceptible to interference, making it difficult to accurately extract systolic duration, diastolic duration, and the cardiac cycle. Thus, ECG is still required as a gold standard. In [20], researchers observed a linear correlation between mechanical time from heart movement and BP values, and used this correlation to estimate BP fluctuations. In [21] and [22], short-time Fourier transform (STFT) was applied to high-frequency radar chest echoes to generate an integral spectrum waveform, from which a U-Net model extracted cardiac intervals for BP estimation. Although this method effectively suppresses respiratory and noise interference through high-frequency processing, it still relies on ECG data for model training and lacks physiological interpretability.

To address these challenges, this work employs FMCW radar to capture chest micro-vibrations induced by cardiac pumping activity. The resulting phase-reconstructed continuous waveform, which precisely maps the mechanical displacement of the chest wall induced by cardiac systole and diastole, is formally defined in this work as the Radar Kinetocardiogram (RKG). Signal quality is enhanced through beamforming and improved differentiate-and-cross-multiply (IDACM) processing, while the novel Adaptive Respiratory-Harmonic Variational Mode Extraction (ARH-VME) algorithm extracts RKG by suppressing respiratory harmonics while preserving critical cardiac signal features. The RKG waveform characteristics are highly correlated with BP formation mechanisms, enabling continuous BP estimation through CTIs. The main contributions are as follows.

- 1) Unlike conventional radar-based BP methods requiring high-fidelity pulse waveforms or ECG-assisted PTT calculation, our approach operates solely with a single FMCW radar to extract RKG signals. By leveraging CTIs temporal markers from the RKG rather than detailed waveform morphology, this method reduces signal quality requirements and enhances robustness in complex environments.
- 2) This work introduces the ARH-VME algorithm, which features adaptive harmonic perception and heartbeat protection mechanisms. This algorithm precisely suppresses respiratory harmonics without breath-holding, extracting high-quality RKG signals whose fiducial points accurately identify cardiac mechanical intervals.
- 3) The method enables high-frequency continuous BP monitoring, generating estimates from 15-s RKG segments. Validation across various respiratory states and

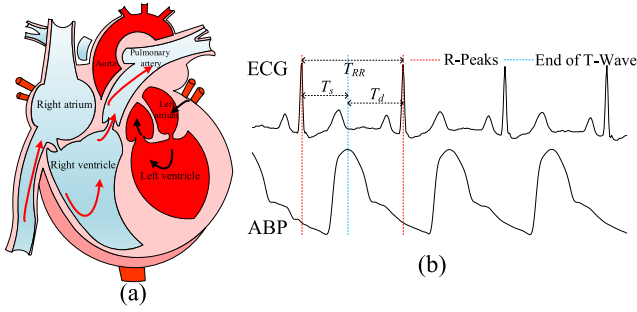


Fig. 1. (a) Schematic plot of cardiac blood circulation. (b) ECG and ABP schematic plot.

40–100 cm sensing distances and radar–chest incidence angles up to 45° within the AAMI-referenced error thresholds. Cross-platform generalizability is further confirmed on an independent public dataset using a different radar modality.

Section II develops a physiologically-grounded BP estimation model using CTIs. Section III presents the ARH-VME algorithm for respiratory harmonic suppression, defines RKG-ECG correspondence, and establishes the BP estimation framework. Section IV validates performance on 25 subjects and an independent public dataset. Section V discusses robustness under varying conditions, computational complexity, parameter sensitivity, and limitations. Section VI concludes this article.

II. BP MEASUREMENT BASED ON CTIs

Accurate BP measurement is crucial for the diagnosis, treatment, and prognosis of cardiovascular diseases. This section reviews the fundamental mechanisms of BP generation and then develops BP estimation using CTIs.

A. BP Generation and Major Influencing Factors

Arterial BP (ABP) results from the combined effects of blood flow pressure against arterial walls and cardiac pumping action. During systole, the left ventricle contracts forcefully, increasing blood flow velocity and PWV. Blood is rapidly ejected into the aorta, causing arterial wall distension and an increase in BP, which reaches the peak arterial pressure, namely systolic BP (SBP). Large arteries are elastic and can store part of the stroke volume (SV) during systole. During diastole, they help maintain pressure through elastic recoil, thereby sustaining blood flow. Small arteries and arterioles constitute the main peripheral vascular resistance (R). During diastole, blood flow velocity decreases along with PWV. Blood then flows through peripheral resistance vessels to other tissues, leading to a progressive pressure decline until arterial pressure reaches its minimum value, namely diastolic BP (DBP), before the next ventricular contraction. BP formation is closely related to cardiac SV, R , arterial compliance (C), and vascular wall elastic modulus (E) [23]. To intuitively illustrate the relationship between cardiac contraction/relaxation and arterial pressure waveforms, Fig. 1(a) presents the cardiac circulation schematic, while Fig. 1(b) demonstrates the

correspondence between ECG and ABP within one cardiac cycle, annotating temporal parameters including T_s , T_d , and T_{RR} , whose physical significance will be further utilized in subsequent modeling.

During cardiac diastole, after aortic valve closure when intravascular blood flow becomes zero, the Windkessel model indicates that aortic pressure decreases with a characteristic decay time. The exponential decay of aortic pressure over time can be expressed as follows [24]:

$$P(t) = P_0 e^{-t/RC} \quad (1)$$

where $P(t)$ represents the aortic pressure at time t , P_0 denotes the initial pressure at the onset of diastole; R represents peripheral vascular resistance, C denotes total arterial compliance, and this model characterizes the dynamic propagation characteristics of BP fluctuations.

In fact, each pressure wave generated by ventricular ejection propagates through arterial vessels at a specific velocity, known as PWV, which significantly influences BP. Under the assumptions of ideal fluid behavior and uniform vessel wall thickness, PWV can be calculated according to the Moens–Korteweg equation as follows [25]:

$$PWV = K \sqrt{\frac{Eh}{2\rho r}} \quad (2)$$

where $E = E_0 e^{\gamma P}$, E_0 is the elastic modulus at zero pressure, γ is a coefficient, P is the BP, K represents the Moens constant. h is the vessel wall thickness; ρ is blood density; and r is the vessel radius. This equation indicates that PWV depends on the physical properties of the vasculature itself. The vascular elastic modulus is not constant but increases exponentially with P . Higher BP results in stiffer vessels and reduced C .

The Bramwell–Hill equation directly relates PWV to hemodynamic parameters. Within a complete cardiac cycle, the PWV is calculated as follows [26]:

$$PWV = \sqrt{\frac{V \cdot \Delta P}{\rho \cdot \Delta V}} \quad (3)$$

where ΔP is the difference between SBP and DBP, namely pulse pressure (PP). The corresponding vessel volume change ΔV represents the total blood volume pumped into the aorta during one cardiac cycle, defined as SV.

Therefore, from (3), the PP equation can be expressed as

$$PP = \frac{\rho}{V} \cdot SV \cdot (PWV)^2. \quad (4)$$

This section has formulated several mathematical models for BP. Many physiological parameters in these models, including C , r , E , and SV, are difficult to measure directly and continuously in practice. Section II-B introduces readily measurable surrogate indices to replace these parameters and construct a practical BP estimation model.

B. BP Model Based on CTIs

Within a single cardiac cycle, the characteristic time intervals associated with ventricular systole and diastole carry rich information about cardiovascular status. The ECG reflects

the heart's electrical activity and is commonly used as the reference for delineating CTIs. The ECG R-wave marks the onset of systole, whereas the onset of diastole coincides with the end of the T wave [27]. As illustrated in Fig. 1(b), the systolic duration is denoted by T_s , the diastolic duration by T_d , and the RR interval by T_{RR} , where $T_{RR} = T_s + T_d$.

Although CTIs are not a direct measure of BP, they are tightly coupled to BP through fundamental physiology. Variations in the durations of systole and diastole reflect the dynamic balance between ventricular pumping function and the mechanical properties of the vascular system, and these two factors jointly determine BP. Therefore, by analyzing CTIs, one can indirectly infer changes in BP.

Over a complete cardiac cycle, and according to (1), the pressure at the end of diastole, namely the DBP, can be obtained as follows:

$$DBP = P_0 e^{-T_d/RC} \quad (5)$$

where DBP is influenced by multiple factors, among which R is key. DBP is approximately proportional to R . From fluid mechanics, R is inversely related to vessel diameter, where larger diameters correspond to lower resistance [28]. During cardiac diastole, let the arterial diameter change be Δd . An increase in Δd reduces R , which in turn decrease DBP. Accordingly, DBP can be expressed as follows [19]:

$$DBP \propto e^{-\alpha \Delta d} \quad (6)$$

where α denotes a proportionality coefficient.

During ventricular systole, a longer T_s allows more ejection time and is associated with a larger SV. The greater volume injected per beat produces stronger arterial wall distension, increasing Δd . Consequently, T_s is approximately proportional to Δd , which yields

$$DBP \propto e^{-\beta T_s} \quad (7)$$

where β denotes a new empirical coefficient that incorporates the original coefficient α and the proportionality between Δd and T_s .

Based on (7), the DBP can then be modeled directly from the T_s as follows:

$$DBP_{est} = DBP_0 e^{-\beta(T_s - T_{s0})} \quad (8)$$

where DBP_0 denotes the baseline of DBP, and T_{s0} denotes the baseline of systolic duration.

From (4), PP and SV are proportional to the PWV. From (2), PWV is strongly influenced by the arterial elastic modulus and is proportional to the E . Therefore, PP satisfies the following proportional relationship:

$$PP \propto SV \cdot E. \quad (9)$$

From the analysis of ventricular systole, T_s is proportional to the SV. When blood vessels become harder and the elastic modulus E increases, the heart rate rises compensatorily, which shortening T_{RR} . Hence, E is inversely related to T_{RR} . The PP can be expressed as [19]

$$PP \propto \frac{T_s^2}{T_{RR}}. \quad (10)$$

According to (10), the calculation models for PP and SBP can be further constructed from the T_{RR} corresponding to RR, as shown in the following equation:

$$PP_{est} = PP_0 \cdot \left(\frac{T_s^2}{T_{RR}} \right) \bigg/ \frac{T_{s0}^2}{T_{RR0}} \quad (11)$$

$$SBP_{est} = DBP_{est} + PP_{est} = DBP_0 e^{-\beta(T_s - T_{s0})} + \left[PP_0 \cdot \left(\frac{T_s^2}{T_{RR}} \right) \bigg/ \frac{T_{s0}^2}{T_{RR0}} \right] \quad (12)$$

where PP_0 is the reference value and T_{RR0} is the reference RR interval time.

This section develops a continuous BP model grounded in CTIs. The model jointly considers three timing parameters and their interrelations, thereby providing a solid theoretical basis for a robust noninvasive BP estimation framework.

III. MEASUREMENT AND PROCESSING OF CHEST MICROMOTION SIGNALS

As described in Section II, CTIs-based BP estimation models provide the theoretical foundation for noninvasive monitoring. However, conventional methods rely on contact ECG sensors, presenting limitations including skin irritation and wearing discomfort, while failing to directly capture SV-related information. This study proposes a radar-based noncontact method for acquiring cardiac mechanical activity by detecting chest wall micro-vibrations. The overall workflow is illustrated in Fig. 2. The radar first captures chest reflection signals, which are then processed by beamforming, phase unwrapping, and respiratory harmonic suppression to obtain the RKG. The RKG is a chest-wall micro-motion signal dominated by cardiac mechanical activity. CTIs are subsequently extracted from the RKG and used as the core features for BP estimation via the RKG-BP model. To address the critical challenge of respiratory interference, this article proposes the ARH-VME algorithm, which adaptively removes the respiratory fundamental and its harmonics while preserving cardiac components.

A. FMCW Radar Fundamentals

FMCW radar transmits continuous wave signals whose frequency changes linearly with time, commonly called chirp signals. The radar operates with one transmit antenna and four receive antennas. The transmission signal is given by

$$X_T(t) = A_T \exp \left(j2\pi \left(f_c t + \frac{B}{2T_c} t^2 \right) \right) \quad (13)$$

where A_T is the amplitude of the transmitted signal, f_c is the starting frequency, B is the signal bandwidth, and T_c is the frequency sweep period.

Assuming an initial distance d_0 between the radar and the subject and a chest micromotion R caused by respiration and heartbeat, and the time delay is given by

$$\tau = \frac{2(d_0 + R)}{c} \quad (14)$$

where c is the speed of light.

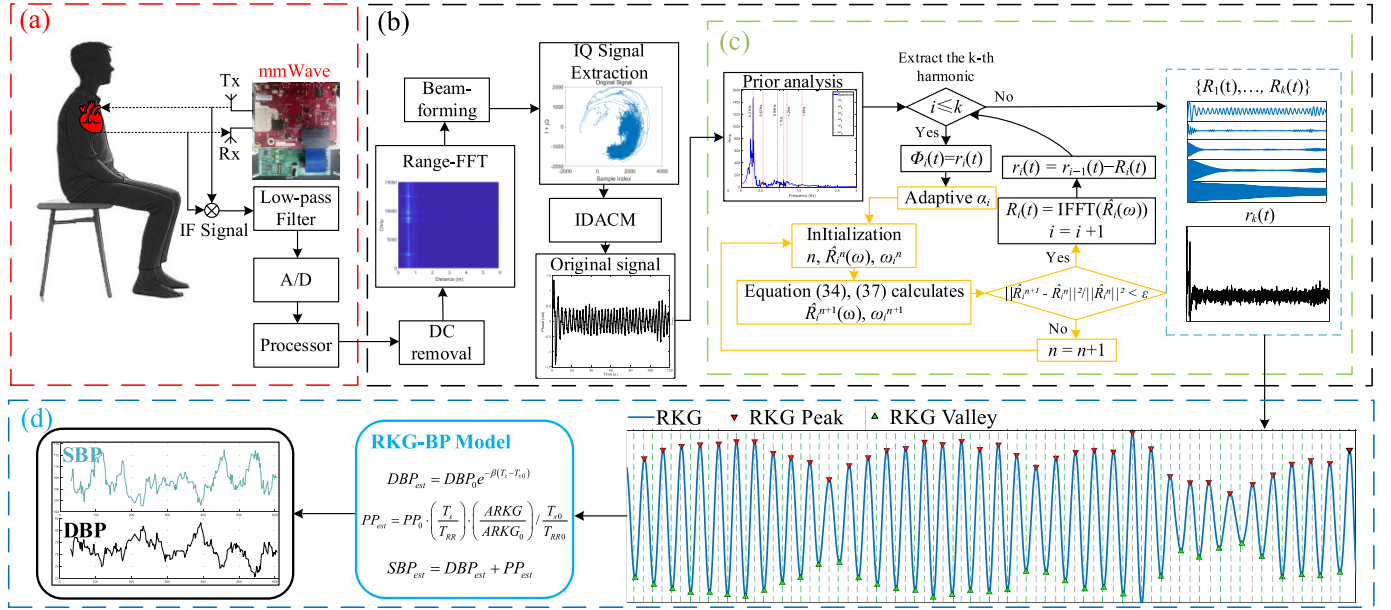


Fig. 2. Overall flowchart. (a) FMCW radar system. (b) Signal process. (c) ARH-VME. (d) RKG-BP.

The echo signal is given by

$$X_R(t) = A_R \exp \left(j2\pi \left(f_c \left(t - \frac{2(d_0 + R)}{c} \right) + \frac{B}{2T_c} \left(t - \frac{2(d_0 + R)}{c} \right)^2 \right) \right) \quad (15)$$

where A_R is the amplitude of echo signal.

In a FMCW system, the received echo is mixed with a the transmit chirp, and the product is passed through a low pass filter. The resulting intermediate frequency signal $H(t)$ carries the range and motion dependent phase term. The high frequency sum term produced by mixing is removed by the low pass filter and is small enough to be ignored. $H(t)$ is shown in the following formula:

$$\begin{aligned} H(t) &= A_T A_R \exp \left\{ j2\pi \left[\left(f_c(t - \tau) + \frac{B}{2T_c}(t - \tau)^2 \right) - \left(f_c t + \frac{B}{2T_c} t^2 \right) \right] \right\} \\ &= A_T A_R \exp \left[-j2\pi \left(\frac{B\tau}{T_c} t + f_c \tau - \frac{B\tau^2}{2T_c} \right) \right] \\ &\approx A \exp \left[-j2\pi \left(\frac{B\tau}{T_c} t + f_c \tau \right) \right] \\ &\approx A \exp \left[-j \left(2\pi \frac{2B(d_0 + R)}{cT_c} t + \frac{4\pi f_c(d_0 + R)}{c} \right) \right]. \quad (16) \end{aligned}$$

With $f_{IF} = ((2B(d_0 + R))/cT_c)$, $\varphi = ((4\pi f_c(d_0 + R))/c)$, the relationships between chest micromotion and the frequency and phase of the IF signal are

$$\Delta f_{IF} = \frac{2BR}{cT_c} \quad (17)$$

$$\Delta \varphi = \frac{4\pi f_c R}{c}. \quad (18)$$

The phase of the IF signal is used to track the chest surface displacement caused by the heartbeat. The RKG waveform

can be reconstructed by processing and analyzing the phase sequence extracted from the radar echo signal.

B. Beamforming for Signal Enhancement

To enhance the SNR and spatial focusing capability of radar echo signals, this study employs a digital beamforming algorithm. Phase signals are obtained through coherent combination of signals from multiple receiving antennas. Since chest respiratory signals and cardiac signals are quasi-periodic, we calculate the SNR autocorrelation coefficients for multiple range-angle combinations and select the combination with maximum autocorrelation coefficient. This approach focuses the array's main beam toward the chest cardiac region while suppressing interference and noise from other directions.

Consider a receiving array composed of N antennas. When a far-field narrowband signal impinges on this array, the signal $M_n(t)$ received by the n th antenna element can be expressed as

$$M_N(t) = A_N e^{j\varphi_N(t)} \quad (19)$$

where $M_n(t)$ is the complex signal received by the n th antenna, A_N denotes the signal amplitude, and $\varphi_N(t)$ is the received phase incorporating both the propagation delay and the phase of the source signal.

To direct the main receiving beam toward the desired angle θ , the weighting coefficient $W_n(\theta)$ is given by

$$W_n(\theta) = e^{-\frac{j2\pi d(n-1)\sin(\theta)}{\lambda}} \quad (20)$$

where d denotes the antenna spacing, λ is the signal wavelength, and n is the antenna index.

For a four-element antenna array ($N = 4$), the beamformed output signal is given by

$$BF(\theta, t) = \sum_{n=1}^4 W_n^*(\theta) M_n(t). \quad (21)$$

This weighted summation enhances signals from the target direction θ through coherent addition while suppressing interference from other directions.

C. Phase Unwrapping

Following the beamforming process, the output signal $BF(\theta, t)$ is a complex signal that encodes the phase information associated with the target's range and angle. For notational simplicity, its slow-time sequence at the human body range bin is denoted as $s(t)$

$$s(t) = A(t)e^{j\phi(t)} \quad (22)$$

where ϕ denotes the total phase incorporating chest displacement from both respiration and cardiac activity.

The chest displacement caused by respiration is significantly larger than that induced by cardiac activity, causing phase variations that often exceed the $[-\pi, \pi]$ range. If arctangent demodulation is directly applied to $s(t)$, the wrapped phase corrupts the weak cardiac signal, making it difficult to recover. To address this, this study employs an IDACM algorithm. The core principle is to reconstruct a continuous unwrapped phase by computing the phase differences between consecutive signals and performing integration. This approach avoids direct calculation of absolute phase, thereby effectively mitigating the phase ambiguity problem caused by large-amplitude respiratory motion.

The instantaneous angular frequency is obtained from $s(t)$ and its time derivative as

$$\frac{d\phi(t)}{dt} = \frac{\text{Im}\left(\frac{ds(t)}{dt} \cdot s(t)^*\right)}{|s(t)|^2} \quad (23)$$

where $\text{Im}(\cdot)$ denotes the imaginary part, and $s(t)^*$ represents the complex conjugate of $s(t)$.

The continuous unwrapped phase $\Phi(t)$ is then reconstructed by integrating the phase rate, as shown in

$$\Phi(t) = \int_0^t \frac{\text{Im}\left(\frac{ds(\tau)}{d\tau} \cdot s(\tau)^*\right)}{|s(\tau)|^2} d\tau + \arg(s(0)). \quad (24)$$

D. RKG-Based CTIs Estimation Using ARH-VME

The extraction of RKG from raw radar echoes proceeds through the four stages illustrated in Fig. 2: beamforming, IDACM-based phase unwrapping, ARH-VME respiratory suppression, and cardiac component extraction. The ARH-VME algorithm, which constitutes the core of this pipeline, is detailed below.

Following the radar signal processing, the weak phase information associated with cardiac activity has been extracted. This section formally defines the RKG, a new noncontact physiological signal that captures the chest-wall mechanical displacement induced by cardiac systole and diastole, obtained after suppressing respiratory fundamental and harmonic components using the ARH-VME algorithm. We then describe its physiological significance, the definition of its characteristic fiducial points, and its corresponding relationship with the ECG.

Radar chest echoes contain complex physiological information. Under normal conditions, heart rate ranges from

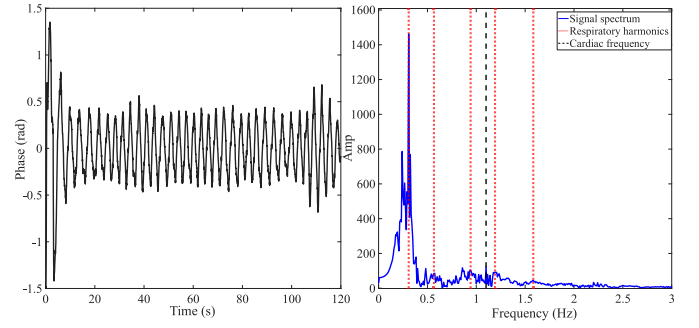


Fig. 3. Position of respiratory fundamental wave and its harmonics in the spectrum.

48 to 120 beats/min, while respiratory rate ranges from 9 to 30 breaths/min. Since respiratory-induced chest displacement substantially exceeds cardiac-induced displacement, the spectral magnitude of respiratory components can be tens to hundreds of times greater than that of cardiac signals, making respiratory interference suppression critical.

As illustrated in Fig. 3, the third, fourth, and fifth respiratory harmonics may overlap with the cardiac frequency band. Consequently, both the fundamental respiratory component and its harmonics must be eliminated. Variational mode decomposition (VMD) requires predefined values for the mode number and regularization parameter, making it challenging to simultaneously separate the broadband fundamental respiratory component and narrowband harmonics while preserving the complete cardiac signal. This often results in under-decomposition or over-decomposition [29]. In clinical applications requiring high-precision CTIs parameters, merely separating the fundamental respiratory component is insufficient. Although these respiratory harmonics possess relatively low energy, they severely distort the fine morphology of RKG, thereby degrading the accuracy of CTIs extraction. Therefore, developing a method that intelligently identifies and precisely removes all respiratory harmonics while ensuring cardiac signal integrity is essential for extracting high-quality RKG signals.

This study employs an ARH-VME algorithm to adaptively remove the fundamental respiratory component and its harmonics while preserving cardiac activity information. The algorithm transforms the complex global decomposition problem into an iterative single-mode extraction process. Unlike conventional VMD, which simultaneously decomposes the signal into a predefined number of modes and is sensitive to this choice, ARH-VME fixes $K = 2$ per iteration, targeting one respiratory harmonic and one residual, and processes harmonics sequentially, thereby avoiding the mode number selection problem. The algorithm adaptively adjusts the regularization parameter to match the spectral characteristics of different harmonics, and incorporates an active protection mechanism to prevent cardiac signal corruption.

Assume the chest echo phase signal $\Phi(t)$ is expressed as

$$\Phi(t) = R(t) + r(t) \quad (25)$$

where $R(t)$ denotes the respiratory signal and $r(t)$ represents the residual signal.

The core task of the algorithm is to precisely extract the target harmonic mode $R_i(t)$ from the current signal $\Phi(t)$. To achieve optimal separation, the ideal mode $R_i(t)$ must simultaneously satisfy three interdependent conditions: First, each component $R_i(t)$ is assumed to be compact around a center pulsation frequency ω_i , which is determined by a prior knowledge. Second, it should maximally capture the target harmonic content from the original signal, thereby minimizing the residual signal energy. Third, it must actively avoid any interference with the a priori cardiac frequency band. To this end, these three objectives are unified into a composite loss function for joint optimization. Specifically, the optimization problem is formulated as follows:

$$\begin{aligned} \min_{R_i} \{ & \alpha_i J_1(R_i, \omega_i) + J_2(R_i, \Phi_i) + \beta J_3(R_i, \omega_h) \} \\ \text{s.t. } & \Phi_i(t) = R_{i+1}(t) + r_{i+1}(t) \end{aligned} \quad (26)$$

where $\Phi_i(t) = r_i(t) = \Phi(t) - \sum_{k=1}^i R_k(t)$, $\Phi_0(t)$ represents the original phase signal $\Phi(t)$, the subscript i of $R_i(t)$ represents the i th harmonic component, and α is the parameter used to balance the bandwidth constraint.

The J_1 term aims to minimize the spectral bandwidth of the extracted mode $R_i(t)$ around its center frequency ω_i , ensuring mode compactness. It is mathematically expressed as the squared norm of the analytic signal gradient

$$J_1(R_i, \omega_i) = \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * R_i(t) \right] e^{-j\omega_i t} \right\|_2^2 \quad (27)$$

where δ denotes the Dirac distribution.

The J_2 term minimizes the energy of the residual signal $r_i(t)$, ensuring that the target mode $R_i(t)$ is maximally extracted from the input signal $\Phi_i(t)$

$$J_2(R_i, \Phi_i) = \|\Phi_i(t) - r_i(t)\|_2^2. \quad (28)$$

The J_3 term prevents the algorithm from erroneously extracting cardiac signals while eliminating respiratory harmonics. This is achieved by penalizing the energy of the extracted mode $R_i(t)$ within the known cardiac frequency band

$$J_3(R_i, \omega_h) = \|H(\omega) \cdot \hat{R}_i(\omega)\|_2^2 \quad (29)$$

where $H(\omega)$ is a Gaussian protection kernel centered at ω_h , ω_h is defined as

$$H(\omega) = \exp \left(-\frac{(\omega - \omega_h)^2}{2\sigma_h^2} \right) \quad (30)$$

where σ_h is the bandwidth set by heartbeat protection.

To update $R_i(t)$, (26) can be expressed as the following equation:

$$\arg \min_{R_i} \{ \alpha_i J_1(R_i, \omega_i) + J_2(R_i, \Phi_i) + \beta J_3(R_i, \omega_h) \}. \quad (31)$$

The optimization problem is transformed from time domain to frequency domain by using Parseval's theorem. The frequency-domain objective function becomes

$$L(\hat{R}_i) = \int_0^\infty [2\alpha_i(\omega - \omega_i)^2 |\hat{R}_i(\omega)|^2 + |\hat{\Phi}_i(\omega) - \hat{R}_i(\omega)|^2 + \beta |H(\omega) \hat{R}_i(\omega)|^2] d\omega. \quad (32)$$

To minimize the integral, minimize the integrand of any frequency ω , and setting the complex derivative to zero yields

$$\begin{aligned} \frac{\partial}{\partial \hat{R}_i(\omega)} (2\alpha_i(\omega - \omega_i)^2 |\hat{R}_i|^2 + |\hat{\Phi}_i - \hat{R}_i|^2 \\ + \beta |H(\omega)|^2 |\hat{R}_i|^2) = 0. \end{aligned} \quad (33)$$

Solving for R_i yields the frequency-domain update

$$\hat{R}_i(\omega) = \frac{\hat{\Phi}_i(\omega)}{1 + 2\alpha_i(\omega - \omega_i)^2 + \beta |H(\omega)|^2}. \quad (34)$$

After the respiratory harmonic mode $\hat{R}_i$ of the current iteration step is updated, the center frequency ω_i is updated. The center frequency ω_i only appears in the bandwidth constraint, so (26) can be shown by the following equation:

$$\arg \min_{\omega_i} \{ \alpha J_1(R_i, \omega_i) \}. \quad (35)$$

When R_i is fixed, J_2 and J_3 are independent of the variable ω_i to be optimized, so the optimization problem is equivalent to minimizing J_1 only, which gives

$$\arg \min_{\omega_i} \int_0^\infty 2\alpha(\omega - \omega_i)^2 |\hat{R}_i(\omega)|^2 d\omega. \quad (36)$$

The center frequency update rule is

$$\omega_i = \frac{\int_0^\infty \omega |\hat{R}_i(\omega)|^2 d\omega}{\int_0^\infty |\hat{R}_i(\omega)|^2 d\omega}. \quad (37)$$

The algorithm iteratively alternates between (34) and (37) to obtain the optimal center frequency. Its performance is primarily influenced by the value of α_i . An excessively large α_i leads to insufficient extraction of the fundamental component, retaining low-frequency residuals that interfere with high-frequency cardiac signals. Conversely, an excessively small α_i may result in overextraction of higher-order harmonics, introducing noise or adjacent frequency components. To address this issue, this article designs an adaptive α_i optimization method based on spectral characteristics to ensure the algorithm employs appropriate α_i values for extracting each respiratory harmonic.

To this end, an adaptive α_i parameter selection strategy based on spectral morphology analysis is developed, utilizing the -3 dB bandwidth to quantify the spectral width of the target harmonic. Starting from the spectral peak of the harmonic to be extracted, a bilateral search is performed until the amplitude drops to half of the peak value. The frequency difference between these two points defines the -3 dB bandwidth $B_{3\text{dB}}$, from which α_i can be derived, yielding

$$\alpha_i = \gamma_i \cdot \frac{C_i}{(B_{3\text{dB}})^{P_i}} \quad (38)$$

where γ_i is the adjustment factor, and C_i and P_i are the coefficients corresponding to the frequency band.

The ARH-VME algorithm requires no external reference signal; all prior spectral knowledge is obtained exclusively from the radar signal itself. The respiratory fundamental frequency is identified by a dual time-frequency verification strategy: a sliding window locates the most periodic segment of the bandpass-filtered phase signal, and the resulting

Algorithm 1 ARH-VME

Input $\Phi_0(t)$
Initialize
 $\{\omega_1^{(0)}, \dots, \omega_N^{(0)}\}, \omega_h \leftarrow FFT[\Phi_0(t)], H(\omega) \leftarrow \omega_h, r_0(t) \leftarrow \Phi_0(t)$
for $i = 1$ **to** N **do**
 $\Phi_i(t) \leftarrow r_{i-1}(t), B_{3dB} \leftarrow (\Phi_i(t), \omega_i^{(0)}), \alpha_i \leftarrow B_{3dB}$ using Eq. (38)
Initialize $\hat{R}_i(\omega), n \leftarrow 0, \omega_i \leftarrow \omega_i^{(0)}$
repeat
 $n \leftarrow n + 1$
update $\hat{R}_i(\omega)$ using Eq (34)
update ω_i using Eq (37)
 $\frac{\|\hat{R}_i^{n+1} - \hat{R}_i^n\|_2}{\|\hat{R}_i^n\|_2} < E$ or $n > N_{\max}$
until convergence:
 $r_i(t) \leftarrow \Phi_i(t) \leftarrow IFFT(\hat{R}_i(\omega))$
end for
Output: $r_K(t), \sum_{k=1}^K R_k(t)$

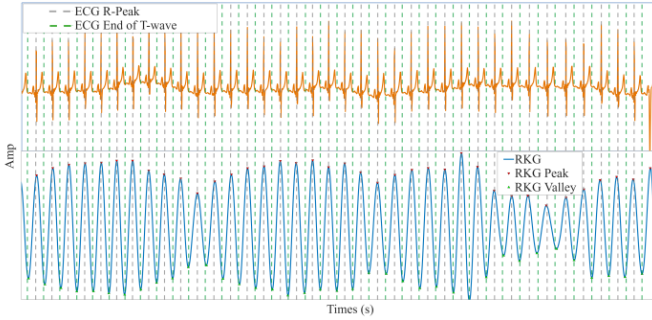


Fig. 4. Corresponding plot of CTIs between RKG and ECG.

inter-peak-interval estimate is refined against the nearest FFT spectral peak. Harmonic frequencies are derived as integer multiples of the fundamental and retained only if their spectral amplitude exceeds a threshold relative to the fundamental. The cardiac frequency is identified as the dominant peak within the physiological heart rate band for the protection kernel. These frequencies serve as initial center frequencies for the iterative modal optimization.

E. RKG-BP Model

After extracting each respiratory harmonic $R_k(t)$, the RKG waveform is obtained by extracting the component centered at frequency ω_h from the residual signal. As illustrated in Fig. 4, following respiratory interference suppression using ARH-VME, RKG extraction focuses on frequencies near the fundamental cardiac frequency to emphasize CTIs characteristics.

The RKG peak corresponds to the moment of peak ventricular contraction, aligning closely with the R-peak of the ECG. Following ventricular depolarization, intraventricular pressure rises rapidly. Shortly after the R-peak, the isovolumetric contraction phase ends as the aortic valve opens, allowing rapid blood ejection from the left ventricle into the aorta. The result-

ing hemodynamic force and anterior chest wall motion are captured by the RKG as its peak mechanical displacement. The RKG trough corresponds to the onset of ventricular diastole, aligning closely with the end of the T-wave. This coincides with the closure of the aortic valve, marking the ventricle's entry into the rapid filling phase. The mechanical vibrations of the chest wall during this phase originate primarily from turbulent blood flow during passive ventricular filling and the elastic recoil of the ventricular wall, which together produce the minimum mechanical displacement captured by the RKG trough. Thus, RKG amplitude variations not only reflect chest wall displacement but are also fundamentally linked to key physiological parameters of cardiac motion.

To quantify CTIs from the RKG, the signal is first smoothed via a moving average filter, after which peaks and troughs are detected using an adaptive amplitude threshold with a minimum inter-peak interval constrained by the plausible heart rate range. The RR interval T_{RR} is computed as the time difference between consecutive peaks, the systolic duration T_s as the peak-to-trough interval, and the diastolic duration T_d as T_{RR} minus T_s . These parameters, together with the RKG amplitude $ARKG$, are fed into the BP estimation model described below.

Established research in Ballistocardiography (BCG) and Seismocardiography (SCG) has confirmed a correlation between the amplitude of these signals—which represent chest wall acceleration—and SV. This implies that chest wall displacement can influence SV [30]. Within each cardiac cycle, the RKG amplitude is primarily determined by the anterior chest wall motion generated during the ventricular ejection phase. This indicates that RKG amplitude correlates with SV and is consequently linked to BP changes. Therefore, parameters extracted directly from RKG can be applied to the CTIs-based BP estimation model described in Section II, enabling fully noncontact BP monitoring. However, traditional cycle-based models primarily rely on temporal parameters and do not fully incorporate the intensity of cardiac mechanical activity. To address this, we modify (11) to obtain the following expression:

$$PP_{est} = PP_0 \cdot \left(\frac{T_s}{T_{RR}} \right) \cdot \left(\frac{ARKG}{ARKG_0} \right) / \frac{T_{s0}}{T_{RR0}} \quad (39)$$

where $ARKG$ is the amplitude of RKG in a cardiac cycle, and $ARKG_0$ is the amplitude reference value. The SBP is then given by

$$\begin{aligned} SBP_{est} &= DBP_{est} + PP_{est} \\ &= DBP_0 e^{-\beta(T_s - T_{s0})} \\ &\quad + \left[PP_0 \cdot \left(\frac{T_s}{T_{RR}} \right) \cdot \left(\frac{ARKG}{ARKG_0} \right) / \frac{T_{s0}}{T_{RR0}} \right]. \end{aligned} \quad (40)$$

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Experimental Equipment and Methods

This study employs the TI AWR1843 radar board with a DCA1000 data acquisition board to collect chest echo data. ECG signals are acquired using the KS1081 evaluation development board from Kingsense. Continuous BP signals are measured using the Finapres Medical System, a clinically

TABLE I
RADAR PARAMETER TABLE

Parameter	Value
Tx	1
Rx	4
Start frequency	77GHz
Bandwidth	3GHz
Frequency slope	50MHz/ μ s
Idle time	15 μ s
Ramp end time	60 μ s
Sample points	64
Sample rate	2MHz
Frame periodicity	5ms

TABLE II
SUBJECT INFORMATION SHEET

Subject	Gender/Age	Height(cm)	Weight(kg)	Health type
1	M/24	168	70	Healthy
2	M/27	178	75	Healthy
3	M/24	173	70	Healthy
4	M/36	179	73	Healthy
5	M/45	172	62	Hypertension
6	M/65	168	70	Hypertension
7	F/50	156	62	Hypertension
8	M/77	178	61.5	Hypertension
9	F/60	152	69	Healthy
10	F/56	157	55	Hypertension
11	M/44	180	99	Hypertension
12	M/56	173	59	Healthy
13	M/31	175	68	Healthy
14	F/28	162	58	Healthy
15	M/52	170	76	Hypertension
16	F/41	165	63	Healthy
17	M/68	172	71	Hypertension
18	F/35	158	54	Healthy
19	M/29	182	80	Healthy
20	F/72	155	60	Hypertension
21	M/54	176	85	Hypertension
22	F/27	164	57	Healthy
23	M/62	174	78	Hypertension
24	M/43	177	75	Healthy
25	F/58	160	66	Hypertension

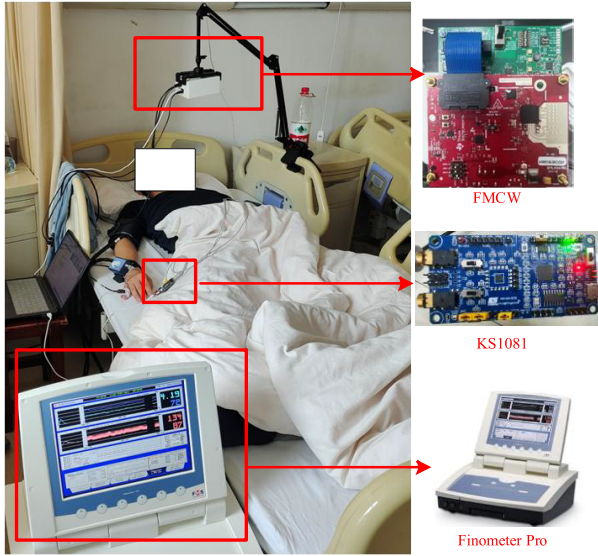


Fig. 5. Experimental equipment and measurement scenarios.

validated noninvasive continuous BP monitor, serving as the reference for BP evaluation in this study. The radar parameters are configured as shown in Table I. The experimental setup is illustrated in Fig. 5. Subjects remain in a supine and resting state during data acquisition, with radar, ECG, and continuous noninvasive BP data collected simultaneously. The radar is positioned 60 cm in front of the subject's chest. Each data acquisition session lasts 5 min, with a 5-min interval between consecutive sessions. Initial calibration is performed by a single cuff-based BP measurement before radar monitoring. The CTIs and RKG amplitude baselines are derived from the first few cardiac cycles of the radar recording. During continuous monitoring, these baseline values are adaptively updated after each estimation window to track gradual physiological changes.

This study conducted clinical data collection and validation on 25 subjects. The participant cohort included patients of various ages and both sexes, with detailed demographics provided in Table II. All experiments were carried out in a clinical setting with the ambient temperature maintained at approximately 25 °C. The cohort consisted of 16 males and

9 females, among whom 12 had been clinically diagnosed with BP abnormalities. With reference to the numerical error criteria commonly used in the AAMI and BHS standards [31], [32], this section evaluates the measurement accuracy of the proposed method. A comparative analysis was performed between the proposed approach and existing relevant methods. Performance was assessed using the following metrics: ME, standard deviation (SD), Mean Absolute Error (MAE), and Root Mean Square Error (RMSE). The expressions for these metrics are given below

$$ME = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i) \quad (41)$$

$$SD = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (42)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (43)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (44)$$

where N is the total number of samples, representing the real value and the predicted value.

B. Signal Processing Results

The raw echo signal from one subject is processed using FFT to obtain the range slow-time map shown in Fig. 6(a), from which the approximate human body range is determined.

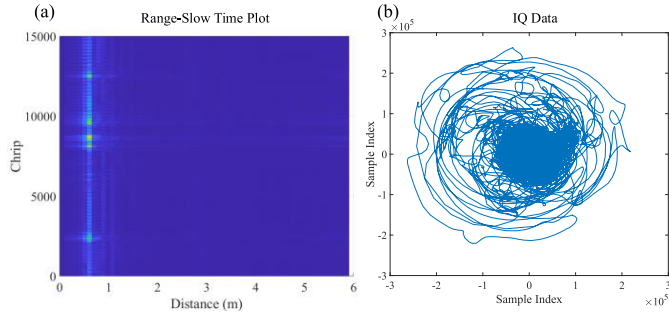


Fig. 6. (a) Radar range slow-time map. (b) Raw radar echo IQ signal from the chest region.

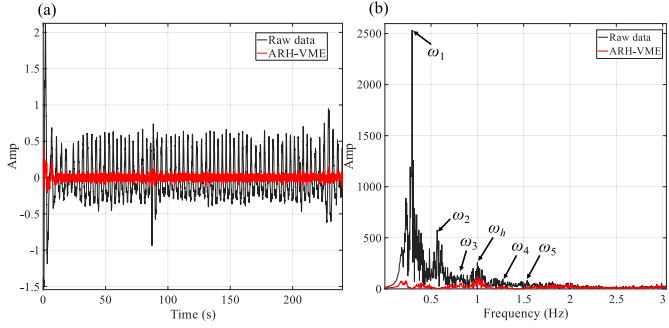


Fig. 7. (a) Original and ARH-VME processed phase signals. (b) Spectra of original and ARH-VME processed phase signals.

A range-angle search is then performed around this approximate position to identify the range-angle combination with the highest autocorrelation coefficient. Digital beamforming is applied to focus on and extract the raw radar echo signal from the chest cardiac motion region. As illustrated in Fig. 6(b), the IQ signal from the chest region is extracted after IDACM processing. The extracted chest phase signal is shown in Fig. 7(a), where the black curve represents the original chest phase signal. It can be observed that large-amplitude low-frequency respiratory oscillations are superimposed on high-frequency cardiac vibrations, with this strong respiratory interference severely masking the weak phase variations induced by cardiac mechanical activity. As illustrated in Fig. 7(b), the spectrum of the original chest phase signal contains the 1st through 5th respiratory harmonics, with the 3rd to 5th harmonics interfering with the cardiac frequency band and compromising cardiac motion information extraction. After applying ARH-VME, the red curve demonstrates that the slow-varying respiratory components are effectively suppressed while the periodic phase characteristics induced by cardiac activity are clearly revealed. The spectrum shows that both the fundamental respiratory component and its harmonics are eliminated, maximally preserving cardiac motion information.

C. RKG Extraction Results and Analysis

To evaluate the algorithm's robustness, subjects sequentially underwent 60 s of resting, 20 s of apnea, and 40 s of Valsalva maneuver. As shown in Fig. 8, for one representative subject during the resting state, CTIs parameters from RKG and

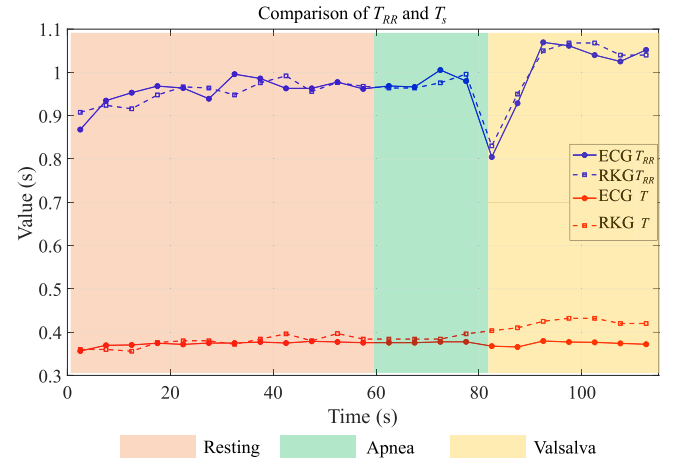


Fig. 8. Comparison of RKG and ECG cardiac interval parameters under different respiratory states.

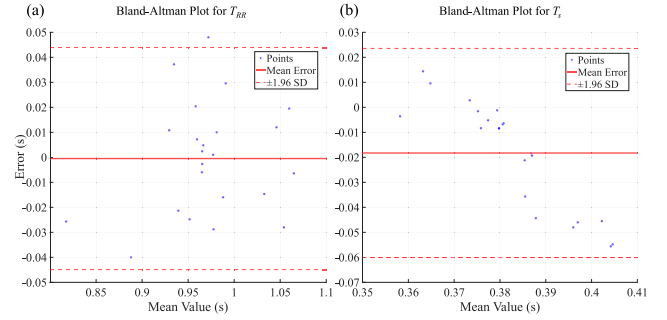


Fig. 9. Bland-Altman plot of RKG and ECG.

ECG exhibit strong agreement. The T_{RR} fluctuates smoothly, while the T_s , represented by the red curve, remains relatively stable during rest. Even in the presence of natural respiration, RKG accurately extracts cardiac characteristic points. During apnea, both T_{RR} and T_s undergo significant physiological changes, which RKG accurately tracks throughout these dynamic processes. Following apnea onset, T_{RR} gradually shortens, consistent with the physiological response to breath-holding. During this phase, RKG-derived T_{RR} maintains strong concordance with ECG-derived T_{RR} , demonstrating the algorithm's robustness during rapid heart rate variations. RKG-derived T_s remains nearly constant during apnea, consistent with ECG-derived T_s , reflecting the relative independence of ventricular systolic duration from heart rate under normal myocardial function. This T_s stability validates RKG's capability to accurately identify landmark moments of cardiac mechanical motion despite heart rate fluctuations. During the Valsalva maneuver, despite complex hemodynamic changes, RKG maintains synchronization with ECG. The deviation in RKG-derived T_{RR} does not increase significantly throughout the Valsalva maneuver, demonstrating the algorithm's adaptability to intense intrathoracic pressure changes and hemodynamic fluctuations. RKG-derived T_s exhibits minor errors with small magnitudes, likely attributed to systematic bias introduced during the Valsalva maneuver.

As shown in the Bland-Altman plots in Fig. 9, RKG demonstrates excellent agreement with ECG in cardiac interval

TABLE III
COMPARISON OF CTIS EXTRACTION ERROR BETWEEN THIS METHOD
AND OTHER METHODS

Method	This work		VMD [29]		TVF-EMD [13]	
	T_{RR}	T_i	T_{RR}	T_i	T_{RR}	T_i
Subject	ME/SD (ms)	ME/SD (ms)	ME/SD (ms)	ME/SD (ms)	ME/SD (ms)	ME/SD (ms)
1	16.03/15.72	-26.65/12.30	95.34/60.25	-128.46/60.25	77.42/41.14	-102.63/7.31
2	22.15/39.86	33.96/34.84	-106.28/58.94	115.67/75.70	-81.45/38.52	96.34/28.12
3	16.28/28.46	-42.55/24.26	89.72/60.76	-118.35/77.89	73.14/39.36	-85.20/44.91
4	18.23/9.44	-23.57/14.46	98.51/63.01	-124.73/81.98	76.86/56.53	-91.10/52.79
5	-5.76/19.90	13.63/19.90	82.16/63.55	132.58/73.04	65.27/51.39	104.31/12.36
6	9.95/18.13	46.88/27.65	-112.64/61.66	145.92/74.50	-86.32/36.04	126.08/56.64
7	15.21/14.60	-26.33/34.86	101.83/62.08	-136.27/77.83	82.60/38.38	103.36/68.68
8	30.13/34.12	-48.20/47.05	126.75/58.55	-152.18/72.23	102.52/51.61	-126.83/30.94
9	27.63/20.88	-37.26/29.99	118.39/61.84	-148.62/72.32	100.13/30.94	-123.09/49.12
10	-26.37/23.47	41.12/19.15	-120.58/62.55	141.36/74.35	-98.46/29.10	125.20/38.85
11	-19.80/43.51	43.90/29.69	-115.27/63.20	138.94/74.41	-96.50/80.41	-120.05/62.66
12	-12.25/23.22	23.74/19.78	103.68/64.58	-129.83/72.23	85.39/61.82	-105.47/61.82
13	8.42/26.87	-31.28/17.55	107.42/63.29	134.25/74.34	91.38/33.28	115.67/27.70
14	-15.73/27.20	25.16/34.08	96.85/60.64	-125.68/76.09	78.52/40.08	-98.34/42.88
15	24.85/29.13	-18.92/27.68	109.54/64.13	-140.57/79.27	88.76/38.70	108.29/125.76
16	-7.38/24.59	39.45/24.19	-113.26/63.28	142.83/72.66	-92.14/42.21	-117.82/51.11
17	19.56/26.80	-22.74/31.59	100.67/62.55	-137.94/77.38	81.23/44.19	121.56/54.41
18	-21.42/34.09	14.38/25.93	117.83/60.90	-130.42/75.67	95.67/52.07	-88.76/69.37
19	11.83/25.31	-35.61/21.84	-97.26/62.91	127.65/73.68	-79.85/50.69	102.34/62.41
20	-14.29/30.68	38.93/37.66	112.48/61.84	-133.71/78.04	96.42/52.40	-104.28/55.97
21	13.72/22.76	-27.41/26.41	105.92/63.45	-139.26/78.05	89.64/34.94	-103.58/76.42
22	-9.84/29.89	32.88/29.42	102.34/63.95	-128.57/72.96	82.47/46.81	-96.31/57.89
23	18.29/21.90	-24.63/31.73	-108.76/63.46	143.68/75.16	-92.15/48.77	-121.74/50.87
24	-16.57/30.77	19.84/36.17	99.18/62.72	-135.24/62.72	87.92/41.82	-109.57/49.69
25	7.91/23.04	-36.02/43.83	93.64/60.12	-131.47/77.00	79.81/47.04	-104.88/50.79

parameters measurements. In Fig. 9(a), the red solid line representing the mean error (ME) lies close to the zero line, indicating no significant systematic bias in the RKG-measured cardiac intervals. For 95 % of the measurements, the differences in cardiac intervals fall within clinically acceptable limits. Although a few data points exhibit slightly larger errors—likely corresponding to moments of pronounced hemodynamic fluctuation during the Valsalva maneuver—all remain within the 95 % limits of agreement. Fig. 9(b) shows that RKG also maintains good agreement with ECG in measuring systolic duration, though a consistent bias is observed. While 95 % of the data points lie within the agreement limits, the stability of the data distribution suggests that this bias is predictable and correctable, thus still meeting clinical requirements. Minor shifts in peak and trough positions are occasionally present in individual RKG cycles. These can be attributed to several main factors: first, the inherent electromechanical delay between the electrical activity captured by ECG and the mechanical contraction reflected in RKG as chest wall displacement; second, the difference in fiducial point definitions, as the ECG T-wave endpoint and the RKG trough mark related but not perfectly simultaneous physiological events; third, transient increases in respiratory interference due to subject movement or deep breathing, which may introduce slight baseline wander in the RKG signal if not fully suppressed. These deviations are consistent across subjects and remain within the 95 % limits of agreement shown in Fig. 9. As summarized in Table III, the proposed method is compared with existing approaches in terms of RKG

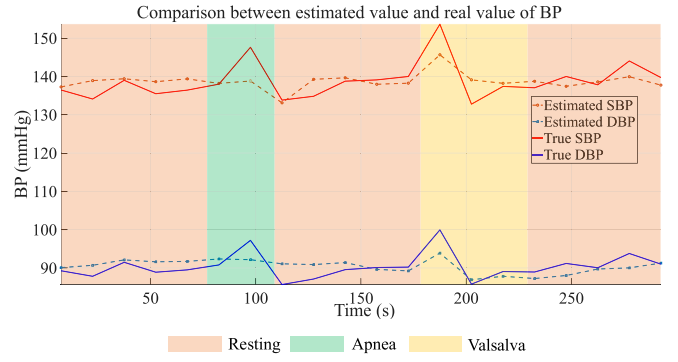


Fig. 10. Comparison between estimated and real value of BP.

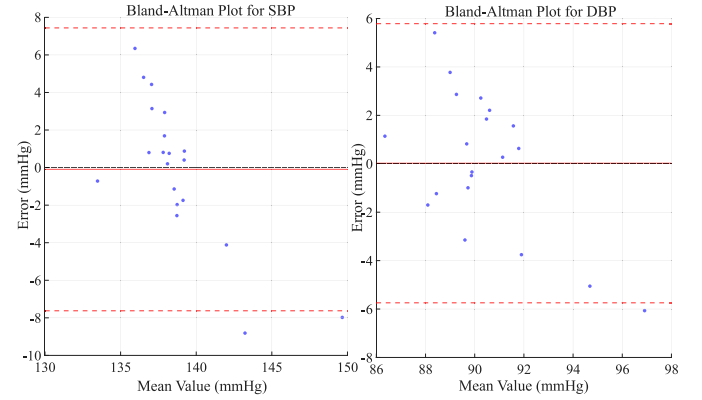


Fig. 11. Bland-Altman Plots for SBP and DBP.

parameter extraction performance, validating the superiority of the presented algorithm.

The correspondence between RKG and ECG fiducial points indicates that radar-measured cardiac-induced chest wall motion can reflect the patterns of mechanical cardiac activity to a certain extent. Overall, RKG can serve as a viable alternative to ECG for extracting key CTIs, enabling noncontact measurement of these parameters. Consequently, once the RKG signal is successfully acquired, the proposed method can be employed to achieve fully noncontact and continuous BP monitoring.

D. Results of Noncontact Continuous BP Monitoring Based on RKG

Fig. 10 shows the noncontact BP measurement results for one representative subject under varying respiratory conditions. The shaded background regions indicate different breathing patterns during the experiment. Across all respiratory segments, the estimated values closely align with the reference values, remaining stable with minimal fluctuations throughout the recording period. To further quantify the accuracy of the method, Bland-Altman analysis was performed, as shown in Fig. 11. For SBP, the majority of data points fall within ± 8 mmHg, with 95 % limits of agreement ranging from -7.7 to 7.7 mmHg. The mean bias is close to zero, indicating negligible systematic error. For DBP, the error distribution is more concentrated, with 95 % limits of agreement

TABLE IV
STATISTICAL INDICATORS OF BP ESTIMATION ERROR FOR EACH SUBJECT

ID	SBP					DBP				
	AAMI	BHS	ME $\pm$ SD	MAE	RMSE	AAMI	BHS	ME $\pm$ SD	MAE	RMSE
1	Yes	A	-1.28 $\pm$ 5.01	4.34	5.14	Yes	A	-0.48 $\pm$ 2.66	2.16	2.66
2	Yes	B	-2.00 $\pm$ 7.82	7.00	7.95	Yes	A	-1.68 $\pm$ 3.95	3.35	4.23
3	Yes	B	-0.98 $\pm$ 6.75	5.31	6.74	Yes	A	-2.56 $\pm$ 3.74	3.62	4.48
4	Yes	A	4.58 $\pm$ 3.77	5.66	5.93	Yes	A	-0.18 $\pm$ 2.81	2.32	2.76
5	Yes	A	2.11 $\pm$ 5.17	4.39	5.58	Yes	A	0.94 $\pm$ 3.53	2.93	3.60
6	Yes	B	-3.48 $\pm$ 7.98	7.37	8.71	Yes	A	1.78 $\pm$ 3.67	3.47	4.03
7	Yes	B	-4.6 $\pm$ 7.73	7.3	9.00	Yes	B	-3.65 $\pm$ 6.68	5.31	7.61
8	Yes	A	1.95 $\pm$ 6.82	4.78	7.09	Yes	A	-1.24 $\pm$ 3.16	2.48	3.39
9	Yes	A	3.23 $\pm$ 5.74	5.65	6.59	Yes	A	1.49 $\pm$ 3.03	2.64	3.33
10	Yes	A	2.98 $\pm$ 5.79	5.35	6.43	Yes	A	0.69 $\pm$ 3.20	2.77	3.22
11	Yes	B	-3.21 $\pm$ 7.45	6.86	8.11	Yes	A	1.36 $\pm$ 3.82	2.86	4.05
12	Yes	A	-1.16 $\pm$ 6.69	5.47	6.79	Yes	A	-2.03 $\pm$ 2.91	2.99	3.51
13	Yes	A	2.43 $\pm$ 5.68	4.92	6.27	Yes	A	-0.73 $\pm$ 2.94	2.13	3.03
14	Yes	B	-4.16 $\pm$ 6.92	6.63	8.07	Yes	A	2.51 $\pm$ 3.58	3.24	4.37
15	Yes	A	0.87 $\pm$ 6.21	4.56	6.28	Yes	A	-1.92 $\pm$ 3.41	2.67	3.91
16	Yes	A	3.64 $\pm$ 6.73	5.81	7.65	Yes	A	2.86 $\pm$ 4.23	3.75	5.11
17	Yes	B	-2.73 $\pm$ 7.18	6.42	7.68	Yes	A	0.58 $\pm$ 3.76	2.41	3.80
18	Yes	A	1.29 $\pm$ 5.95	4.68	6.11	Yes	A	-2.45 $\pm$ 3.89	3.18	4.60
19	Yes	A	-1.82 $\pm$ 5.64	5.23	5.93	Yes	A	1.73 $\pm$ 2.85	2.56	3.33
20	Yes	B	4.91 $\pm$ 7.38	6.95	8.86	Yes	B	-4.23 $\pm$ 5.91	5.07	7.27
21	Yes	A	1.12 $\pm$ 6.05	5.02	6.18	Yes	A	-0.72 $\pm$ 3.05	2.41	3.13
22	Yes	B	-2.36 $\pm$ 7.12	6.02	7.56	Yes	A	1.08 $\pm$ 2.95	2.36	3.04
23	Yes	A	3.04 $\pm$ 6.38	5.63	7.18	Yes	A	-1.54 $\pm$ 3.18	2.75	3.53
24	Yes	A	-1.89 $\pm$ 5.76	4.68	6.06	Yes	A	-2.31 $\pm$ 3.42	3.11	4.13
25	Yes	B	4.12 $\pm$ 7.38	6.97	8.59	Yes	A	0.64 $\pm$ 3.56	2.84	3.62

between -5.9 and 5.9 mmHg, and most points lying within ± 4 mmHg, demonstrating higher measurement precision. No evident trending bias is observed in either SBP or DBP, confirming that the measurement error remains consistent across different pressure levels and respiratory conditions without systematic drift. These results validate the accuracy and reliability of the proposed noncontact BP monitoring approach, demonstrating its robust overall performance even in the presence of respiratory variations.

Table IV lists in detail the statistical indicators of BP estimation error for each subject. For SBP, the error statistics of all 25 subjects fall within the numerical limits commonly referenced from the AAMI standard, although several do not reach the BHS grade-A threshold. For DBP, the error statistics of all subjects fall within the AAMI-referenced limits, and most also reach the BHS grade-A threshold. This indicates that the method achieves good accuracy and reliability at the individual level, although several subjects still exhibit relatively large errors, likely due to motion artifacts or intersubject physiological differences during measurement.

To further validate the proposed method, we compared it with several recently published radar-based BP monitoring methods, as summarized in Table V. The proposed method achieves competitive performance across all evaluation metrics. Compared with other deep learning-based methods, it provides comparable SBP and DBP accuracy, while retaining explicit physiological interpretability by modeling BP from RKG-derived CTIs rather than relying on an entirely end-to-end black-box network. The model only requires an initial calibration for each subject. In comparison with methods that do not use deep learning, which also require

subject-specific calibration, the proposed approach achieves similar or better accuracy with a single chest radar sensor, without multisensor fusion or strict dependence on fine-grained waveform morphology. As a result, it can meet the accuracy requirements under normal spontaneous breathing, offering more relaxed constraints on subject posture and breathing and thus improved comfort. BP is estimated on a beat-to-beat basis, with each cardiac cycle yielding one SBP and one DBP value. Error statistics in Table IV are computed from nonoverlapping 15-s segment averages between the radar estimates and the Finapres reference. Specifically, the group-level ME $\pm$ SD in Table V is aggregated from the 25 per-subject ME values rather than pooled across all estimates, the ME $\pm$ SD of SBP is $0.3 (5.72) \pm 6.44$ mmHg, and the ME $\pm$ SD of DBP is $-0.4 (3.02) \pm 3.6$ mmHg. These results fall within the numerical error limits commonly referenced from the AAMI standard, and also reach the BHS grade-A threshold. Overall, Table V demonstrates that the combination of RKG-CTIs and the ARH-VME filtering strategy enables a favorable tradeoff between accuracy, sensor configuration, interpretability, and user comfort compared with existing radar-based BP methods.

To assess the statistical significance of the estimation performance, a one-sample t-test was conducted on the per-subject ME values from all 25 subjects to test for systematic bias. For SBP, the t-statistic is 0.448 with a p-value of 0.658; for DBP, the t-statistic is -1.045 with a p-value of 0.306. Neither result reaches the 0.05 significance level, confirming the absence of systematic estimation bias. The 95 % confidence intervals for the group-level metrics are as follows: SBP ME of 0.26 $[-0.95, 1.48]$ mmHg, MAE of 5.72 $[5.32, 6.12]$ mmHg, and SD of 6.44 $[6.03, 6.86]$ mmHg; DBP ME of $-0.40 [-1.20, 0.39]$ mmHg, MAE of 3.02 $[2.69, 3.34]$ mmHg, and SD of 3.60 $[3.22, 3.97]$ mmHg. The ME intervals for both SBP and DBP encompass zero, consistent with the t-test results. The upper bound of the DBP SD interval remains at 3.97 mmHg, well below the 8 mmHg threshold commonly referenced from the AAMI standard.

E. Validation on a Public Dataset

To evaluate the generalizability of the proposed method, additional validation was conducted on the publicly available radar vital signs dataset by Schellenberger et al. [33], which contains synchronized CW radar, ECG, and continuous ABP signals from 30 subjects recorded in a clinical environment. Since the CW radar lacks range resolution, the beamforming stage was omitted and the ARH-VME algorithm was applied directly to the single-channel phase signal.

The group-level CTIs extraction results across all 30 subjects yield a T_{RR} ME of $+14.9$ ms with SD of 136.0 ms, and a T_s ME of $+43.5$ ms with SD of 66.8 ms. The error distributions are shown in Fig. 12. For BP estimation, the SBP ME $\pm$ SD is -1.7 ± 5.0 mmHg with MAE of 7.26 mmHg, and the DBP ME $\pm$ SD is -1.7 ± 3.9 mmHg with MAE of 5.26 mmHg. Both SBP and DBP fall within the numerical error limits commonly referenced from the AAMI standard at the group level. A one-sample t-test confirms no statistically significant systematic bias for either SBP or DBP, with p-values of 0.252 and 0.115,

TABLE V
COMPARISON WITH OTHER RADAR-BASED BP METHODS

Method	Position	Features	DL	Filtering Method	Accuracy	
					SBP	DBP
ECG	Chest	ECG-CTIs	No	No	(MAE) $\pm$ SD (mmHg) (6.46) $\pm$ 7.23	(MAE) $\pm$ SD (mmHg) (3.95) $\pm$ 5.36
HSMM-Unet [22]	Chest	Integrated spectrum-CTIs	Yes	BF	(MAE) $\pm$ SD (mmHg) (4.84) $\pm$ 7.51	(MAE) $\pm$ SD (mmHg) (3.82) $\pm$ 5.82
PTT [34]	Wrist /Chest	Displacement	No	No	RMSE (mmHg) 3.60	RMSE (mmHg) 4.38
rPTT [7]	Wrist	PW-rPTT	No	IIR	ME $\pm$ SD (mmHg) 0.55 $\pm$ 5.45	ME $\pm$ SD (mmHg) -2.26 $\pm$ 3.93
CaPTT [9]	Neck/Chest	PW-RDT	No	BF	ME $\pm$ SD (mmHg) 5.54 $\pm$ 7.62	ME $\pm$ SD (mmHg) 4.68 $\pm$ 6.15
nPTT [35]	Wrist /Chest	PW-nPTT	Yes	EMD	(MAE) $\pm$ SD (mmHg) 9.51 $\pm$ 12.13	(MAE) $\pm$ SD (mmHg) 6.69 $\pm$ 8.12
DWP [10]	Wrist	PW-DWP	No	ARHF	ME $\pm$ SD (mmHg) 3.78 $\pm$ 9.02	ME $\pm$ SD (mmHg) 1.27 $\pm$ 3.76
RSD-NET [11]	Chest	CPMV	Yes	BF	ME(MAE) $\pm$ SD (mmHg) -0.32(4.86) $\pm$ 6.14	ME(MAE) $\pm$ SD (mmHg) -0.2(4.42) $\pm$ 5.5
airBP [14]	Wrist	Waveform	Yes	No	ME $\pm$ SD (mmHg) -0.3 $\pm$ 4.8	ME $\pm$ SD (mmHg) -0.23 $\pm$ 3.79
mmBP+ [36]	Wrist	PW	Yes	PSF-FLAF	ME $\pm$ SD(mmHg) -0.65 $\pm$ 3.92	ME $\pm$ SD (mmHg) 1.31 $\pm$ 3.99
DSFNN-BP [13]	Neck	PW	Yes	TVF-EMD	ME(MAE) $\pm$ SD(mmHg) -0.49(4.16) $\pm$ 5.19	ME(MAE) $\pm$ SD (mmHg) -0.32(5.11) $\pm$ 3.87
This work	Chest	RKG-CTIs	No	ARH-VME	ME(MAE) $\pm$ SD(mmHg) 0.3(5.72) $\pm$ 6.44	ME(MAE) $\pm$ SD (mmHg) -0.4(3.02) $\pm$ 3.6

DL: Deep Learning; PW: Pulse Wave; BF: Band-pass filter; IIR: Infinite-Impulse-Response filter;

EMD: Empirical Mode Decomposition; ARHF: Adaptive Respiratory-Harmonic Filtering;

PSF-FLAF: Periodic Signal Feature-based Functional Link Adaptive Filter

respectively. Compared with the in-house FMCW results in Table IV, the higher MAE is attributable to the lower input SNR of the CW radar and the presence of motion artifacts in the less controlled clinical recordings. These results confirm that the ARH-VME algorithm and the CTIs-based BP model generalize across different radar platforms and data acquisition conditions.

V. DISCUSSION

The robustness of a BP measurement method is a key factor for its successful application in clinical practice, and its stability and adaptability under different measurement conditions still require further verification. The factors that affect the robustness of the system include sensor placement, measurement distance, etc., which are directly related to the practicability of the method and the feasibility of clinical deployment.

A. Comparison of Radar Measurement at Different Body Sites

In noncontact physiological signal monitoring, sensor placement directly influences system practicality, user comfort,

and clinical feasibility. In existing studies, radar sensors are commonly employed to monitor different body sites such as the neck, wrist, or chest. Compared to neck and wrist measurements, selecting the chest as the primary monitoring region in this study offers significant advantages.

From a signal quality perspective, the chest region provides a larger surface area, offering an extended effective monitoring range and more flexible sensor positioning options. Although the neck and wrist have superficial blood vessels with relatively minimal respiratory interference, their narrow monitoring areas impose stringent requirements for precise sensor positioning, where slight positional deviations can significantly degrade signal quality. In contrast, chest-based measurements allow sensor placement within a considerable range, reducing dependence on precise positioning and enhancing system robustness. From a user experience and comfort perspective, chest monitoring demonstrates clear advantages in terms of unobtrusiveness. The neck and wrist are highly active and sensitive body regions, particularly in sleep monitoring scenarios where these sites are more susceptible to positional changes and interference. Conversely, the chest region exhibits relatively limited movement,

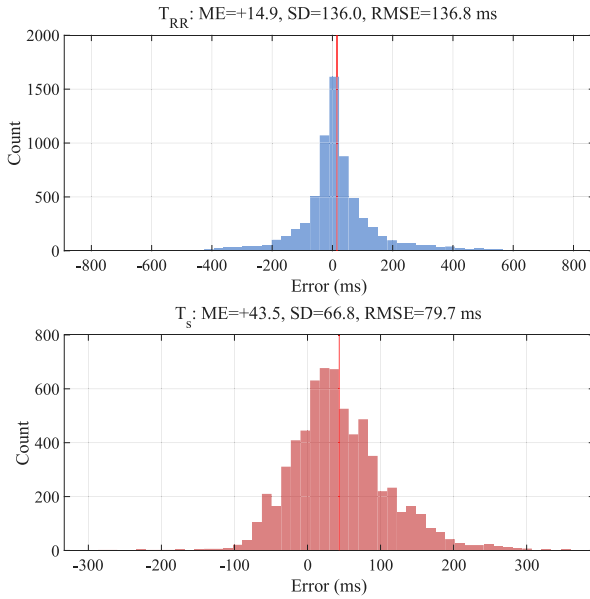


Fig. 12. Error distribution of CTIs extraction on the public dataset. Top: T_{RR} error histogram; Bottom: T_s error histogram. Red vertical lines indicate the ME.

facilitating prolonged continuous monitoring with greater user acceptance.

From clinical application and device integration perspectives, chest measurements present broader application potential. In clinical environments, patients are typically positioned supine or semi-recumbent with the chest region exposed and easily accessible, allowing sensors to be conveniently placed bedside or integrated into hospital bed systems without requiring additional wearables or postural adjustments from patients. This not only streamlines clinical workflows but also reduces dependence on healthcare personnel. Furthermore, in daily living scenarios, chest-based sensors can be more naturally integrated into smart home devices, enabling truly unobtrusive, noncontact health monitoring.

In summary, selecting the chest as the target region for radar monitoring offers advantages not only in signal acquisition stability and robustness but also demonstrates significant application value in user comfort, clinical practicality, and device integration potential, making it an ideal choice for noncontact continuous BP monitoring.

B. Impact of Measurement Distance

In practical applications, the distance between the radar sensor and the subject varies depending on environmental constraints and device installation configurations. Variations in measurement distance directly affect radar echo signal strength, SNR, and phase demodulation accuracy, thereby influencing BP estimation accuracy. To evaluate the robustness and applicability of the proposed method across different measurement distances, this section presents a multidistance measurement experiment that systematically analyzes the impact of distance variation on BP estimation performance. Four measurement distances were examined: 40, 60, 80, and 100 cm.

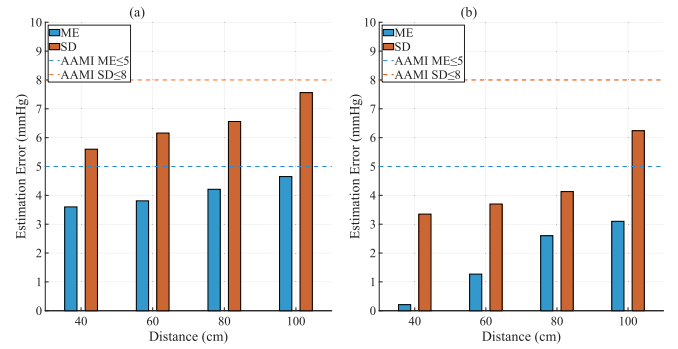


Fig. 13. Accuracy plot of (a) SBP performance and (b) DBP performance at different distances.

The results are shown in Fig. 13. Across all tested distances, both SBP and DBP ME and SD fell within the numerical limits commonly referenced from the AAMI standard, though SBP exhibited larger errors than DBP. These results validate the clinical applicability of the proposed method across the 40–100 cm distance range, with the extended effective range providing important guidance for practical deployment. The RKG method demonstrates excellent distance adaptability, with its core advantage residing in the feature extraction mechanism. Unlike conventional pulse wave methods requiring capture of fine arterial pulsation morphology at the micrometer scale, RKG extracts CTIs by identifying only coarse-grained features such as periodic peaks and troughs, without relying on waveform details susceptible to noise interference. Consequently, even at 100 cm with low SNR, primary characteristic points remain reliably identifiable, ensuring measurement accuracy stability.

C. Impact of Measurement Angle

In practical applications, the radar sensor and the subject's chest are rarely kept perfectly orthogonal. Variations in the measurement angle change the effective projection of chest motion onto the radar line of sight and may also affect the echo strength and SNR, thereby influencing BP estimation accuracy. To evaluate the robustness of the proposed method to angular misalignment, this section presents a measurement-angle experiment that systematically analyzes the impact of different incident angles on BP estimation performance. To assess the robustness of the proposed method under different viewing angles, a multiangle experiment was conducted at 0°, 15°, 30°, 45°, and 60°.

Fig. 14 illustrates the ME and SD of SBP and DBP at different measurement angles. For SBP, as the angle increases, the ME remains below 5 mmHg. When the included angle reaches 45°, the SD value is much lower than 8 mmHg, and when the included angle reaches 60°, the SD value approaches the 8 mmHg threshold commonly referenced from the AAMI standard. For DBP, ME and SD were significantly lower than the AAMI-referenced threshold at all angles.

Within an angular range of up to 45°, the BP estimation errors for both SBP and DBP fall within the AAMI-referenced numerical limits, and have good margin. At 60°, RKG extraction is noticeably degraded. Overall, the results suggest that

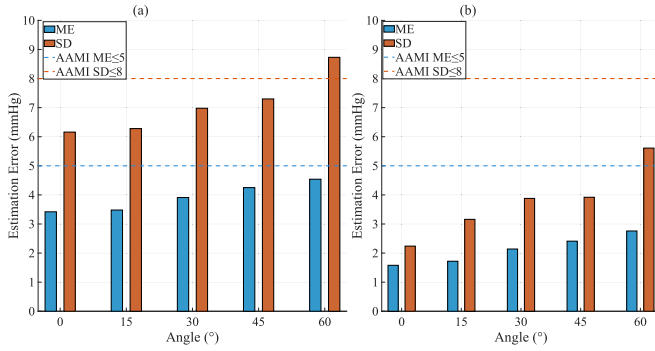


Fig. 14. Accuracy plot of (a) SBP performance and (b) DBP performance at different angles.

TABLE VI
THEORETICAL COMPLEXITY COMPARISON

Method	Complexity	Memory
TVF-EMD [13]	$O(K_i \cdot N_s \cdot (N \log N + N_k^3))$	$O(K_i \cdot N)$
VMD [29]	$O(K_v \cdot N_v \cdot N \log N)$	$O(K \cdot N)$
ARH-VME	$O((M+1) \cdot K_{iter} \cdot N \log N)$	$O(N)$

TABLE VII
RUNTIME COMPARISON ACROSS SIGNAL DURATIONS

Duration	VMD (s)	TVF-EMD (s)	ARH-VME (s)
15s	0.368 ± 0.015	1.102 ± 0.017	0.104 ± 0.002
30s	0.656 ± 0.008	1.771 ± 0.021	0.225 ± 0.009
60s	1.287 ± 0.027	3.630 ± 0.113	0.455 ± 0.014
120s	2.467 ± 0.054	7.678 ± 0.044	1.330 ± 0.033
240s	4.989 ± 0.024	11.348 ± 0.034	2.053 ± 0.007

extreme oblique angles lead to reduced radial motion and stronger clutter, making CTIs extraction more challenging. Therefore, the system does not require the subject to face the radar perfectly orthogonally, allowing natural posture and small torso rotations during normal breathing without significantly compromising measurement accuracy.

D. Computational Complexity

Table VI summarizes the theoretical complexity of the three respiratory suppression methods, where N is signal length; M is number of respiratory harmonics; K_{iter} is convergence iterations per harmonic; K_v and N_v are VMD mode number and iterations; K_i and N_s are number of IMFs and sifting iterations; N_k is number of B-spline knots.

ARH-VME fixes $K = 2$ per iteration and processes each harmonic sequentially, yielding a compact $O(N)$ memory footprint. Table VII compares the empirical processing time on real radar data across five signal durations.

Under the 15-s operating window used in this work, ARH-VME requires 0.104 s, $3.5\times$ faster than VMD and $10.6\times$ faster than TVF-EMD. This advantage is maintained across all tested durations, with real-time factors consistently above $90\times$. Combined with the superior CTIs extraction accuracy shown in Table III, ARH-VME offers a favorable tradeoff between speed and precision for real-time noncontact BP monitoring.

TABLE VIII
PARAMETER SENSITIVITY ANALYSIS ON THE PUBLIC DATASET

Parameter	Value	T_{RR} ME $\pm$ SD (ms)	T_s ME $\pm$ SD (ms)
σ (Hz)	0.05	11.27 ± 99.04	47.45 ± 42.95
	0.10	15.57 ± 94.99	49.83 ± 41.35
	0.20	14.46 ± 95.59	49.44 ± 43.30
M	2	20.85 ± 96.94	52.62 ± 42.99
	5	15.57 ± 94.99	49.83 ± 41.35
	6	20.70 ± 95.65	52.04 ± 39.64
Threshold	0.01	15.75 ± 95.42	49.92 ± 41.53
	0.02	15.57 ± 94.99	49.83 ± 41.35
	0.10	21.00 ± 93.22	52.47 ± 41.21

TABLE IX
ABLATION STUDY ON THE PUBLIC DATASET

Configuration	T_{RR} ME $\pm$ SD (ms)	T_s ME $\pm$ SD (ms)
Full ARH-VME	15.57 ± 94.99	49.83 ± 41.35
No cardiac protection	-0.96 ± 112.62	41.16 ± 51.71
Fixed α	-6.19 ± 101.60	37.91 ± 46.31
Fixed M = 3	5.60 ± 93.26	44.78 ± 41.87

E. Parameter Sensitivity and Ablation Analysis

To evaluate the robustness of ARH-VME to its pre-set parameters, we conducted a sensitivity analysis on the independent public dataset by [33], which provides a cross-platform and lower-SNR test setting than our in-house FMCW data. Three key parameters were independently swept while keeping all others at their default values: the cardiac protection kernel width σ , the maximum harmonic order M, and the harmonic detection amplitude threshold. The detailed results are summarized in Table VIII. Across all three parameter sweeps, the CTIs extraction errors remained stable. The T_s SD varied by less than 2.0 ms across σ , 3.4 ms across M, and only 0.4 ms across the threshold range of 0.01–0.05. The T_{RR} SD similarly remained within a 6 ms range for all three parameters. These results confirm that the default values used in this work fall within a broad region of stable performance rather than representing cherry-picked optima.

An ablation study on the same dataset further quantified the contribution of each design component, as reported in Table IX. Removing the cardiac protection kernel produced the largest degradation, with T_{RR} SD and T_s SD increasing by 18.6 % and 25.1 %, respectively. This confirms that the protection mechanism is essential for preventing cardiac energy from being erroneously extracted during respiratory suppression. Replacing the adaptive α with a fixed schedule also degraded performance, validating the spectral-morphology-based bandwidth adaptation. Fixing M = 3 yielded results comparable to the full configuration, suggesting that the default upper bound mainly provides a safety margin. Together with the comparisons against VMD and TVF-EMD in Table III, these results support both the robustness and the practical effectiveness of ARH-VME.

F. Limitations of This Work

A limitation of the BP model is its inherent sensitivity to timing errors in the extracted CTIs. Since the model is a nonlinear function of T_s and T_{RR} , small extraction errors propagate into BP estimates in a nontrivial manner. A first-order perturbation analysis under representative physiological values shows that a ± 10 ms error in T_s produces approximately ± 0.75 mmHg in DBP and ± 2.6 mmHg in PP, while a ± 10 ms error in T_{RR} produces approximately ± 0.53 mmHg in PP, with the worst case first-order SBP perturbation reaching approximately ± 3.9 mmHg. These bounds may be attenuated in practice by beat-to-beat stochastic averaging and subject-specific baseline calibration, but they underscore the importance of accurate CTIs extraction, which motivates the design of the cardiac protection kernel and adaptive bandwidth control in ARH-VME.

Although the proposed RKG-based noncontact BP monitoring method has achieved promising experimental results, several limitations remain to be addressed in future work. Similar to PTT and rPTT models, the RKG-based BP model is fundamentally a relative measurement approach and requires initial calibration of absolute BP values. The current system also exhibits limited robustness against motion artifacts. While the ARH-VME algorithm effectively separates respiratory components, it struggles to fully eliminate nonperiodic motion interference, such as random body movements, thereby restricting the system's practicality in daily activity scenarios. Furthermore, this study involved 25 subjects, covering a range of ages and both sexes, yet the sample size remains relatively small. The dataset lacks samples from hypotensive patients, which may limit the model's generalizability in extreme BP ranges. Moreover, the Finapres reference used in this study is a noninvasive device rather than the invasive arterial catheterization gold standard. The reported error statistics therefore reflect agreement with the Finapres reference rather than absolute accuracy against intra-arterial pressure, and future validation against invasive arterial measurements in clinically indicated patients is warranted. The influence of factors such as temperature drift and environmental variations in the radar system on long-term measurement stability also requires further in-depth investigation. Although the online adaptive updating mechanism helps mitigate short-term calibration drift, the current 5-min recording duration is insufficient to assess long-term calibration stability over hours or days. Determining the optimal recalibration interval requires dedicated long-duration experiments, which is an important direction for future work.

A limitation of the present study is that the experiments were conducted under relatively controlled conditions, with subjects instructed to remain as still as possible. This design reflects a deliberate proof-of-concept strategy, because the primary objective of this work is to establish and validate the RKG-CTIs-BP pathway, with respiratory harmonic suppression as the specific technical focus. Confirming the feasibility of each stage under well-characterized conditions is a necessary first step before extending the method to more challenging scenarios involving unconstrained body motion and complex environmental interference.

To overcome these limitations, future work will focus on expanding the clinical dataset, developing motion artifact suppression algorithms tailored to real-world scenarios, and exploring deep-learning-based strategies to move toward calibration-free BP estimation. Integrating learning-based motion compensation with physics-informed signal extraction represents a promising hybrid architecture for extending noncontact BP monitoring to more demanding clinical environments. Long-term stability under realistic deployment conditions will be another critical direction.

VI. CONCLUSION

This study introduces a novel continuous noncontact BP estimation technique that first uses an FMCW radar to acquire a RKG from chest-wall microvibrations, and then extracts CTIs from the RKG. Unlike many radar-based BP estimation approaches, the proposed method does not require any ECG prior, relying solely on the radar-derived RKG signal for feature extraction and BP estimation. The proposed ARH-VME algorithm effectively separates respiratory components, achieving superior signal quality compared to conventional filtering methods. Experimental validation involving 25 subjects demonstrated an ME $\pm$ SD of $0.3 (5.72) \pm 6.44$ mmHg for SBP and $-0.4 (3.02) \pm 3.6$ mmHg for DBP, with group-level errors falling within the AAMI-referenced numerical limits for all participants. The system exhibited excellent distance robustness, maintaining clinical-grade accuracy across a measurement range of 40–100 cm. Even at the maximum distance, the ME remained below 5 mmHg. Multiangle experiments further showed that for radar–chest incidence angles up to 45° , both SBP and DBP estimates remained within the AAMI-referenced numerical limits, indicating that the system tolerates moderate angular misalignment and natural torso rotations, while performance degradation becomes evident only at extreme oblique angles. This strong performance stems from RKG's reliance on coarse-grained features rather than fine-grained waveform morphology, endowing it with greater inherent resistance to noise and signal attenuation compared to traditional pulse wave-based methods. By incorporating RKG amplitude information into a new BP model, estimation accuracy was further improved. Although initial BP calibration is still required and motion artifact handling needs enhancement, the proposed method represents a significant step toward practical noninvasive continuous BP monitoring. Future work will focus on developing calibration-free deep learning models that integrate the advantages of RKG, along with clinical validation across diverse populations to improve generalizability.

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